# Mahatsachan or Mirage? Null-Matched Auditing of Minimal Affective Prompt Stimuli in Multilingual LLMs

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***Abstract*—A recurring claim in prompt engineering is that appending a short affective phrase—"Let's think step by step," "This is very important to my career"—reliably improves large language model (LLM) performance. Replication work has cast doubt on these gains, yet the field lacks a decisive control: are the observed shifts caused by the *affective semantics* of the stimulus, or would *any* surface-matched perturbation produce shifts of the same size? We introduce MAPS (Minimal Affective Prompt Stimuli), an auditing protocol that calibrates each candidate stimulus against an empirical null distribution of semantically inert, surface-matched decoys. As a probe we use one high-valence Thai word, มหัศจรรย์ (*mahatsachan*, 'miraculous'), which lets us cross *stimulus language* with *task language*—a manipulation monolingual studies cannot perform. Across 5 models from 3 families and two parallel Thai/English task pairs, under 34 conditions of which $K{=}23$ are matched nulls, the null distribution spans 2.18 accuracy points—as large as the gains typically reported. None of the 10 stimuli, ours or the literature's, exceeds that envelope in the direction of a gain, while pre-registered equivalence tests certify 6 of them—the target included, at $\Delta = +0.00$ points—as *practically inert* at a $\pm 1.5$-point threshold. The one stimulus distinguishable from the nulls runs the other way: the Thai translation of a widely used emotional prompt costs 2.34 points, and the cost traces to a language switch rather than to affect—it makes one model answer English tasks in Thai. A false positive is also caught: the target improves answer-format adherence at $p < .001$, yet the matched nulls improve it as much. A leave-one-null-out check confirms the calibrated statistic is conservative on real data. We release the stimulus bank, harness and per-item outputs.**

***Index Terms*—prompt engineering, large language models, cross-lingual evaluation, Thai NLP, affective computing, reproducibility, statistical significance testing**

## I. Introduction

Few results in applied NLP have travelled as fast as the observation that appending "Let's think step by step" to a prompt substantially improves zero-shot reasoning [1]. The finding was quickly generalised into a folk methodology: a growing inventory of short appended phrases is claimed to unlock latent capability. Emotional stimuli such as "This is very important to my career" are reported to improve performance by wide margins [2]; automatically discovered instructions outperform human-written ones [3]; and system messages invoking *Star Trek* beat careful hand-tuning on GSM8K [4].

This literature has a structural weakness. Reported effects are frequently computed over the *best* stimulus in a set rather than the average, which inflates headline numbers substantially: the widely cited 115% improvement attributed to EmotionPrompt corresponds to roughly 2.6% when averaged across all stimuli in the same tables, and re-testing the same techniques finds "a general lack of statistically significant differences across nearly all techniques tested" [5]. This sits alongside broader evidence that benchmark scores are highly sensitive to semantically irrelevant surface choices—formatting [6], option order [7], minor paraphrase or typos [8]—with spreads of several accuracy points from perturbations carrying no meaning at all.

Putting these two literatures side by side exposes the missing experiment. If meaningless perturbations move accuracy by several points and affective stimuli move it by a few, then **the reported effect of affective prompting has never been shown to exceed the noise floor of arbitrary prompt perturbation**. Existing studies compare against the *empty* baseline; the correct comparison is a *matched-null* baseline—same length, script and attachment, no affective content (Fig. 1). Almost no published work runs this control.

**Our probe.** We instantiate the audit with a deliberately minimal stimulus: the single Thai word มหัศจรรย์ (*mahatsachan*), roughly 'miraculous'. It is *minimal*—one word, no instruction to reason, so no smuggled-in task decomposition; *high-valence*, permitting a contrast against neutral Thai words; and *non-English*, which lets us cross stimulus language with task language—a manipulation monolingual work cannot perform, and which work on politeness suggests may matter [9].

**Contributions.** (i) MAPS, a null-matched auditing protocol, and the *null-calibrated effect size* (NCES), which measures an effect in units of prompt-perturbation noise and whose calibration

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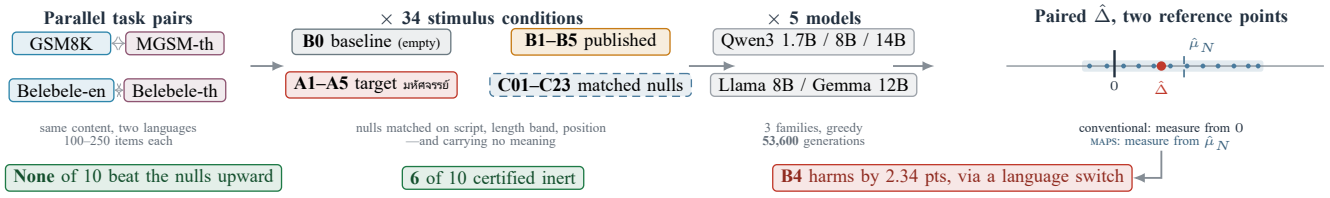



Fig. 1. **The study in one view.** Every condition runs on identical items, so each stimulus yields a paired effect $\hat{\Delta}$ against the baseline. Prior work tests $\hat{\Delta}$ against zero, but the matched nulls form an envelope centred at $\hat{\mu}_N$, not at zero. NCES measures from $\hat{\mu}_N$: does this stimulus do anything a meaningless string of the same shape would not?

we verify on real data (Section III). (ii) A Thai stimulus bank with $K{=}23$ inert decoys matched to the target on script, length band and position (Section IV). (iii) An audit over 34 conditions and 53,600 generations spanning 3 model families up to 14B and two parallel Thai/English task pairs, whose pre-registered equivalence tests deliver a positive claim of *no* effect (Sections V–VI). (iv) A power argument showing the field is underpowered by roughly an order of magnitude, and a recipe for using the protocol in practice.

## II. Related Work

**Instructional and affective stimuli.** Chain-of-thought prompting elicits intermediate reasoning through exemplars [10]; a single appended instruction suffices zero-shot [1]. Automated search over phrasings [3] beats human-written prompts, though the discovered strings are often semantically opaque—a hint that surface form rather than meaning carries part of the effect. EmotionPrompt reports large gains from appended emotional stimuli [2]; Battle and Gollapudi find similar effects from "positive thinking" system messages, but that the best message is model-dependent and sometimes *no message at all* [4]. Sprague et al. show CoT-style gains concentrate on mathematical and symbolic tasks [11].

**Brittleness and replication.** Sclar et al. demonstrate accuracy spreads of up to 76 points from semantics-preserving formatting choices alone [6], and Salinas and Morstatter document comparable instability under minor alterations [8]. Vaugrante et al. re-test several techniques including EmotionPrompt and find no significant differences [5]. We supply the missing *control* rather than another replication, following standard practice on significance testing [12], [13] and using equivalence testing [14] so that nulls are informative.

**Cross-lingual prompting.** MGSM introduced multilingual chain-of-thought and includes Thai [15]; cross-lingual thought prompting [16] shows that instruction language matters independently of task language. Yin et al. find the optimal politeness level is language-dependent [9]—the closest prior work to ours, though without null controls. Thai LLM evaluation is anchored by Typhoon and ThaiExam [17]. To our knowledge no prior study calibrates prompt-stimulus effects against an empirical null distribution of matched inert stimuli.

## III. The MAPS Protocol

### A. Null calibration

For a stimulus $s$ appended to a task template, write $\hat{\Delta}(s)$ for its accuracy change against the unstimulated baseline. Prior work reports $\hat{\Delta}$ and tests it against zero. That test is misspecified: $\hat{\Delta} \neq 0$ is expected even for strings with no semantic content at all.

Let $N = \{n_1, \ldots, n_K\}$ be a bank of *matched null stimuli*—strings matched to the target as a population on surface properties (script, attachment position, and token-count and length band; Section IV tests whether the residual spread matters) but carrying no affective or instructional content. Evaluating over $N$ yields an empirical null distribution $\{\hat{\Delta}(n_k)\}_{k=1}^{K}$ with mean $\hat{\mu}_N$. The machinery is deliberately standard: (3) is the Monte-Carlo permutation $p$-value with the $+1$ correction [18], $\hat{\sigma}^2_{\text{true}}$ the truncated method-of-moments heterogeneity estimator [19], and (2) its prediction interval [20]. The contribution is not the estimator but the object it is applied to—a constructed, surface-matched null population for prompt stimuli, which the prompting literature does not build.

**The spread must be decomposed.** The observed variance of the null effects has two components,

$$\operatorname{Var}[\hat{\Delta}(n_k)] \;=\; \underbrace{\sigma^2_{\text{true}}}_{\text{stimulus-to-stimulus}} \;+\; \underbrace{\mathbb{E}\Big[\operatorname{SE}^2(\hat{\Delta}(n_k))\Big]}_{\text{finite-sample noise}}, \qquad (1)$$

and only the first is a property of the stimulus population. Using the raw standard deviation as a denominator would make "null-calibrated" depend on how many items the *nulls* happened to be run on—not a property of the stimulus under test at all. We therefore estimate $\hat{\sigma}^2_{\text{true}} = \max(0, \operatorname{Var}[\hat{\Delta}(n_k)] - \overline{\operatorname{SE}^2})$ and define the **null-calibrated effect size** as a prediction-interval statistic,

$$\text{NCES}(s) \;=\; \frac{\hat{\Delta}(s) - \hat{\mu}_N}{\sqrt{\operatorname{SE}^2(\hat{\Delta}(s)) + \hat{\sigma}^2_{\text{true}} + \operatorname{Var}[\hat{\mu}_N]}}, \qquad (2)$$

which asks the right question: is the target's effect larger than a *newly drawn* inert stimulus would plausibly produce? Since $\hat{\sigma}^2_{\text{true}}$ is clipped at zero it can collapse when the nulls differ only by sampling noise, making the test anti-conservative; we therefore also compute a conservative variant retaining the undecomposed spread, and require any claim to clear both. We additionally report the rank-based (exact) calibrated $p$-value,

two-sided. Writing $R^{\pm}(s) = 1 + |\{k : \hat{\Delta}(n_k) \gtrless \hat{\Delta}(s)\}|$ for the two tail counts,

$$p_{\text{rank}}(s) \;=\; \frac{2\min\big(R^{+}(s),\, R^{-}(s)\big)}{1+K}, \tag{3}$$

*as a bound, not a test*: its smallest attainable value is $2/(K{+}1) = 0.083$, so across 10 corrected targets it cannot fall below 0.83 whatever the data show. Gating a conclusion on it would make that conclusion arithmetic rather than empirical, so discrimination is carried by the conservative parametric statistic; enlarging $K$ is the only remedy and its cost is linear. We treat the observed null spread $\hat{\sigma}_{\text{obs}}$ as of independent interest: it measures the prompt-perturbation noise floor of a model–task pair, the scale against which any published prompting gain should be read.

### B. Inferential protocol

All conditions are evaluated on byte-identical items within each model–task cell, giving a fully paired design. Each $\Delta$ receives a paired bootstrap over items ($10^4$ resamples, BCa intervals [12], [21]) and an exact McNemar test on per-item correctness pairs—exact rather than $\chi^2$ because discordant counts are small precisely when conditions resemble one another, the regime the null bank inhabits. Evidence is pooled by concatenating the paired item-level contrasts of every eligible cell, each against its own cell's baseline, and bootstrapping over the pooled pairs; this is where the statistical power lives (see below). Because an item recurs under every model, we re-ran the bootstrap resampling *items*, with all of an item's pairs kept together: interval widths change by at most a factor of 1.03 (median 1.00), so treating pairs as independent costs nothing here. Family-wise error is controlled by Holm–Bonferroni [22] *separately within each family*—the baseline tests and the null-calibrated tests—since treating them as one family would be anti-conservative for the first and over-conservative for the second. Finally we report equivalence tests (TOST [14]) against a pre-registered smallest effect size of interest of $\pm 1.5$ accuracy points; rejecting both one-sided tests licenses the positive conclusion that a stimulus is *practically inert*, which mere non-significance never does. The margin, like the stimulus bank, items, seeds, primary outcome and analysis plan, was fixed before any model was queried, and it is deliberately strict: it sits well below the 2.6-point average gain the EmotionPrompt tables imply once every stimulus is counted [5], so an effect of the size the literature reports can never be certified inert, and below the typical $\hat{\sigma}_{\text{obs}}$ (Section VI), so "inert" means smaller than what swapping in a meaningless string already does. Any single margin is a judgement, so we also report the verdicts at $\pm 1.0$ and $\pm 2.0$ (Section VI-E).

**Is NCES calibrated?** Its denominator assumes the null effects behave like draws from one population, so we test that assumption on the data rather than asserting it. Leaving each null out in turn and scoring it as a "target" against the remaining $K{-}1$ gives $K$ stimuli whose true effect is, by construction, that of an inert string. Pooled, 0 of 23 are rejected at $\alpha = .05$ (largest $|\text{NCES}| = 1.19$); per cell, 4 of 460 (0.9%). The conservative statistic thus errs on the side of failing to detect effects, not inventing them, which is the direction a debunking claim can afford.

TABLE I

STIMULUS CONDITIONS, ALL APPENDED AS A SUFFIX. $|t|$ IS TOKEN COUNT UNDER THE REFERENCE (QWEN3) TOKENIZER. VALENCE NORMS WERE NOT ADMINISTERED (SECTION VII). THE NULL BANK HAS $K = 23$ MEMBERS.

| ID | Stimulus | $\lvert t\rvert$ | Val. |
|---|---|---|---|
| *Group A: target and variants* | | | |
| A1 | มหัศจรรย์ *mahatsachan: miraculous, ...* | 8 | — |
| A2 | ทำออกมาให้มหัศจรรย์ *'make it magnificent'* | 11 | — |
| A3 | "mahatsachan" *romanised transliteration* | 4 | — |
| A4 | "magnificent" *English near-gloss* | 3 | — |
| A5 | อัศจรรย์ *atsachan: wondrous* | 7 | — |
| *Group C: matched nulls (lexical controls)* | | | |
| C1 | ธรรมดา *thammada: ordinary, mundane* | 1 | — |
| C2 | น่าเบื่อ *na-bua: boring, tedious* | 5 | — |
| C3 | โต๊ะ *to: table* | 4 | — |
| *Group C: generated null bank* | | | |
| C04–C13 | 10 Thai pseudo-words, non-lexical | 7–11 | — |
| C14–C23 | 10 neutral concrete nouns | 4–9 | — |

**Power, and why the pooled analysis is primary.** Under a paired binomial (McNemar) model, required sample size is driven harder by the *discordance rate*—how often two conditions disagree on the same item—than by the effect size. At the observed 5.2% discordance, detecting $\delta = 2$ points at 80% power needs $n \approx 1{,}028$ paired items; inverting, a 200-item benchmark resolves only $\approx 4.5$ points and a 500-item one $\approx 2.9$. Reported gains from affective prompting are routinely smaller than that, on evaluations of exactly that size—an independent explanation, alongside stimulus cherry-picking, for why such results replicate poorly [5]. We do not escape this per cell and do not pretend to: cell results are descriptive, and inference runs on the pooled design, where 3,200 paired observations per stimulus give a minimum detectable effect of 1.1 points.

## IV. STIMULI

The condition set has three groups: Group A, the target with its morphological and orthographic variants; Group B, literature comparators, which situates the measurement against published techniques on identical items; and Group C, the contribution. Table I shows Groups A and C, whose orthography and surface matching cannot be recovered from the text. Group B holds the two English phrases quoted in Section I (B1, B3), their Thai translations (B2, B4) and OPRO's "Take a deep breath and work on this problem step-by-step" [3] (B5); every stimulus appears with its effect in Table III. C1 (ธรรมดา, *thammada*, 'ordinary') is a valence contrast—same word class and register, opposite valence—so $\Delta(\text{A1}) - \Delta(\text{C1})$ isolates valence from lexical insertion. The generated bank mixes orthographically legal Thai pseudo-words, holding script and length constant while removing meaning entirely, with frequency- and length-matched neutral nouns.

**How tight is the match?** The bank is matched as a population, not member by member: the generated nulls span 1–11 tokens under the reference (Qwen3) tokenizer, median 7 against the target's 8, and the lexical controls C1–C3 are matched on word class and script rather than length. Individual members therefore differ from the target, so we test whether those differences matter. They do not: across the $K$ nulls, pooled $\Delta$ is unrelated to token count (Spearman $\rho = +0.16$, permutation $p = .48$) or character length ($\rho = -0.24$, $p = .28$), and within cells, after centring each cell's nulls, $\rho = -0.01$ ($p = .76$). Real nouns and pseudo-words shift accuracy alike ($-0.05$ vs. +0.34 points). The envelope is not an artefact of its own length spread.

**Valence norming.** "Affective" ought to be measured, not assumed. A blind Self-Assessment Manikin [23] instrument (shuffled per rater, seeded with pseudo-words) is released but was not administered, so valence here rests on lexical semantics (Section VII).

## V. Experimental Setup

**Models.** We use 5 models from 3 families, 1.7B to 14B, varying the two things that could each explain away a null. A *within-family scale ladder* (Qwen3 1.7B, 8B, 14B) holds architecture, tokenizer and corpus fixed while varying parameter count eightfold; *cross-family* models (Llama-3.1-8B, Gemma-3-12B) test whether the result is a quirk of one vendor. Neither alone suffices—the ladder cannot rule out a house style, a cross-family contrast alone confounds vendor with scale. Inference is local through Ollama, greedy ($T$=0, removing sampling variance from every contrast), fixed seed. Two models were screened out on baseline behaviour only, never on a stimulus contrast: Qwen3-0.6B (at the accuracy floor) and Qwen3-4B (a later reasoning-only release our suppression does not reach). Every analysed cell carries its own null bank, run on the same machine as its baseline. This matters: an earlier partial run of Gemma-3-12B on different hardware and a different build of the same model, with identical prompts and greedy decoding, gives a different final answer on 4.1% of 4,433 shared items—a perturbation with no semantic content, of the same order as the stimulus effects under test.

Reasoning is disabled and treated as a controlled factor. This required bypassing the inference server's reasoning switch, whose chat template appends the control token *after* the user content, and so after our suffix stimulus, displacing it from final position. Since attachment position is manipulated, we construct the chat markup directly so the stimulus is last.

**Tasks.** Two *parallel pairs*, each varying task language with content fixed: GSM8K [24] / MGSM-th [15] (arithmetic; MGSM-th is a human translation of a GSM8K subset) and Belebele [25] English / Thai (reading comprehension, the same passages translated). Parallelism is deliberate—comparing ThaiExam against MMLU would confound language with difficulty and item provenance—and two pairs of different reasoning type separate a language effect from a task-type effect. Items are sampled under a fixed seed and are identical across conditions within a model.

TABLE II
Target effect $\Delta_{A1}$ (points) per model and task, with the matched-null noise floor $\hat{\sigma}_{obs}$ in parentheses. Cells are descriptive; claims rest on the pooled tests of Table III.

| Model | Bel-en | Bel-th | GSM | MGSM |
|---|---|---|---|---|
| Qwen3-1.7B | +0.8 (2.9) | +0.4 (2.4) | −4.0 (2.5) | −8.0 (4.5) |
| Qwen3-8B | −1.2 (1.0) | +1.2 (1.7) | +0.0 (1.8) | −1.0 (2.4) |
| Qwen3-14B | +0.0 (1.3) | +1.3 (0.5) | +0.0 (1.2) | +0.7 (3.2) |
| Llama-3.1-8B | +4.0 (2.6) | −5.3 (3.2) | +5.3 (3.9) | +2.7 (4.3) |
| Gemma-3-12B | −2.0 (1.2) | +0.7 (2.0) | +0.0 (1.2) | −0.7 (1.8) |

**Sample sizes.** Groups A and B run on 100–250 items per model–task cell; the $K$ nulls run on a nested 40-item subset of the same items—the regime the decomposition of (1) was built for, since fewer items inflate each null's SE, which is then subtracted rather than absorbed. Pooled over the 20 cells, each main stimulus has 3,200 paired observations and each null 800.

**Secondary measures.** For every generation we log output token count, chain-of-thought emission, answer-format adherence, refusal, and *language drift*—whether a Thai stimulus causes a model to answer an English task in Thai.

## VI. Results

### A. The noise floor

The headline quantity in Table II is $\hat{\sigma}_{obs}$, the spread of effects produced by stimuli that mean *nothing*: across cells it ranges from 0.52 to 4.54 accuracy points, median 2.18. Published affective-prompting gains are of the same order or smaller—EmotionPrompt averages 2.6% once all its stimuli are counted [5]—so a one-stimulus, one-benchmark comparison against an empty baseline cannot separate its effect from this spread. By (1), $\hat{\sigma}_{true}$ is 0.00 in 18 of 20 cells: the nulls are statistically interchangeable, and the whole observed spread is finite-sample noise—which is exactly what a practitioner meets on swapping in a meaningless variant. Since $\hat{\sigma}_{true}$ then clips to zero, every claim uses the conservative denominator.

**The noise floor tracks headroom, not parameter count.** Across cells $\hat{\sigma}_{obs}$ falls as baseline accuracy rises ($r = -0.73$). Scale does not order it: Llama-3.1-8B carries the *largest* mean floor in the study (3.48), above even Qwen3-1.7B, and Gemma-3-12B the smallest (1.54); a single-family ladder would have read the trend as a scale effect. Such effects are thus easiest to demonstrate exactly where they matter least.

### B. Is the target stimulus distinguishable from noise?

Table III gives the pooled tests. Under Holm correction, the zero-effect null is rejected for 1 of the 10 non-null stimuli, and 1 survives null calibration. Direction is what matters: *no stimulus—ours or the literature's—exceeds the matched-null envelope in the direction of a gain.* The affective-prompting claim is not merely unreplicated; there is nothing above the noise floor to replicate.

The target A1 obtains $\Delta = +0.00$ points (95% CI [−0.81, 0.75], $p_{Holm} = 1.000$) and NCES $= -0.17$, on 3,200 paired

TABLE III

POOLED EFFECTS OVER ALL ELIGIBLE CELLS. $p_{\text{Holm}}$ TESTS AGAINST THE EMPTY BASELINE, $p_{\text{null}}$ AGAINST THE MATCHED-NULL BANK, EACH HOLM-CORRECTED WITHIN ITS FAMILY; NCES AND $p_{\text{null}}$ USE THE CONSERVATIVE DENOMINATOR. TOST: PRACTICALLY INERT AT $\pm 1.5$ POINTS.

| ID | $\Delta$ | 95% CI | NCES | $p_{\text{Holm}}$ | $p_{\text{null}}$ | TOST |
|---|---|---|---|---|---|---|
| B1 | +0.97 | [−0.1, 1.9] | +1.08 | 1.000 | 1.000 | – |
| B5 | +0.44 | [−0.6, 1.4] | +0.40 | 1.000 | 1.000 | inert |
| A5 | +0.28 | [−0.6, 1.0] | +0.22 | 1.000 | 1.000 | inert |
| A3 | +0.06 | [−0.7, 0.8] | −0.08 | 1.000 | 1.000 | inert |
| A1 | +0.00 | [−0.8, 0.8] | −0.17 | 1.000 | 1.000 | inert |
| A4 | −0.03 | [−0.8, 0.7] | −0.21 | 1.000 | 1.000 | inert |
| B3 | −0.09 | [−0.9, 0.7] | −0.29 | 1.000 | 1.000 | inert |
| A2 | −1.25 | [−2.2, −0.3] | −1.78 | .396 | .604 | – |
| B2 | −1.56 | [−2.8, −0.4] | −1.96 | .396 | .453 | – |
| B4 | −2.34 | [−3.3, −1.4] | −3.20 | <.001 | .014 | – |

observations; its interval lies wholly inside the pre-registered SESOI, so the target is not merely undetected but certified practically inert (Section VI-E).

**The one calibrated effect runs the wrong way.** The single stimulus that clears both tests is B4, the Thai translation of EmotionPrompt's career-stakes stimulus [2], and it *costs* 2.34 points (95% CI [−3.3, −1.4], NCES $= -3.20$, $p_{\text{null}} = .014$), while its English original is inert on the same items ($-0.09$, [−0.9, 0.7]). Decomposing the pooled estimate locates the cost. In 2 cells (Llama-3.1-8B on Belebele-en and GSM8K) the Thai stimulus switches the *output* language on 100% of English-task generations, and accuracy changes by −16.3 points ([−21.7, −11.3]); in the other 18 cells a residual of $-0.9$ ([−1.9, −0.1]) remains, which the nulls pooled over the same cells match (NCES $= -1.35$, $p_{\text{null}} = .178$). The only effect the nulls cannot explain is thus a model-specific language switch, not an affective response—and the opposite of the gain its proponents report. Without a null bank it would be indistinguishable from perturbation noise; without the crossing it would not be seen at all.

**Where the per-cell variation comes from.** Across the 20 cells of Table II the target runs from −8.00 to +5.33 points, and each cell is the evidence a single-model, single-benchmark study would report. That spread is what sampling predicts. Scoring every main stimulus in every cell against that cell's own envelope, 12 of 200 tests exceed it (6.0%, against 5% expected under no effect); the target exceeds it in 1 of 20. Per model, the target pools to between −1.3 and +1.7 points (5 of 5 intervals cover zero); per task language, to +0.38 ([−0.69, 1.31]) in English and −0.38 ([−1.62, 0.81]) in Thai; per reasoning type, to $-0.08$ ([−1.62, 1.31]) on arithmetic and +0.05 ([−0.89, 0.89]) on comprehension. Pooling does not mask a subgroup effect; no subgroup has one to mask.

### C. Valence and language

The valence contrast A1 vs. C1 holds word class, register and script constant and varies only affective polarity; it gives $\Delta_{\text{A1}} - \Delta_{\text{C1}} = +0.00$ points: changing the polarity of the word changes nothing. Crossing stimulus language with task language over both pairs shows no interaction either: a Thai stimulus gives $-0.67$ points on a Thai task against $-1.27$ on an English one, an English stimulus $-0.11$ and $-0.23$. The Thai-on-Thai cell, which a lexical-semantic account predicts should stand out, does not; every cell lies inside the median noise floor.

### D. A false positive, caught

The sharpest demonstration of what null calibration buys is not on accuracy but on *format adherence*—whether the model emitted the requested answer line. Appending the target improves adherence by +1.12 points (95% CI [0.5, 1.7], exact McNemar $p < .001$) on 3,200 paired items, concentrated where there is headroom: +4.31 points on Thai arithmetic ($p < .001$), whose baseline adherence is only 88.3%, and nil on the comprehension pair, already at 100.0%. Against an empty baseline this is a publishable positive finding.

The story does not survive the control. The 23 matched nulls improve adherence by +1.18 points on average ($-0.75$ to +2.38); against that distribution the target obtains NCES $= -0.08$, $p_{\text{null}} = 1.000$, and 0 of 10 stimuli survive. *Appending anything makes the model attend to the format instruction.* The effect is real, but it belongs to perturbing the prompt, not to what the perturbation means—and no amount of further data against the empty baseline would reveal that.

The target's other logged side effects are nil (drift 0.0% against 0.0% at baseline; refusals +0.00 pts). Across the whole Thai-script bank, Llama-3.1-8B answers English tasks *in Thai* on 22.0% of generations, against 0.3% for every other model—the mechanism behind B4 above, and a hazard invisible to accuracy-only evaluation.

### E. Equivalence

Non-significance is not evidence of absence. For 6 of 10 stimuli—the target among them—TOST rejects both one-sided tests against the pre-registered $\pm 1.5$-point SESOI: not a failure to detect an effect, but a demonstration that an effect large enough to matter is ruled out. For the target the verdict is equivalent, on [−0.81, 0.75] points around +0.00. The verdicts are not an artefact of the chosen margin: at $\pm 1.0$ points 5 stimuli remain certified inert, and the target is inert down to a margin of 0.66 points; at $\pm 2.0$, 7 are. The remaining 4 stay outside equivalence, bounded rather than resolved: 3 on the harmful side (B4, B2 and the target variant A2, of which only B4 clears null calibration) and 1 in the direction of a gain (B1, [−0.1, 1.9]), which does not.

## VII. DISCUSSION

**Using MAPS in practice.** When $\hat{\sigma}_{\text{obs}}$ is of the same order as the reported gains, a one-stimulus, one-benchmark comparison cannot tell an effect from a draw from the perturbation distribution, so NCES or an equivalent calibrated statistic should be a condition of claiming one. The protocol slots into existing pipelines as a gate rather than a new benchmark. (i) For each candidate prompt edit, generate $K \geq 19$ surface-matched nulls—same script, length band and position; the released generator does this for Thai. (ii) Run them on a nested subset

of the evaluation items; at 40 items per null this adds $40K$ generations per model–task, here about half the main budget, and the bank is reusable across candidates of the same shape. (iii) In a prompt optimiser (e.g. OPRO-style search [3]), accept a candidate only if its gain clears the conservative NCES, not merely zero; otherwise the search selects the luckiest perturbation. (iv) In multilingual leaderboards, publish $\hat{\sigma}_{\text{obs}}$ per model–task and treat differences inside it as ties, and log output language alongside accuracy.

**Why minimal cues do so little.** One account is that such cues receive too little attention during encoding to matter. Our evidence is behavioural, but it constrains that account: a single Thai word is processed enough to switch a model's output language, so the cue is not ignored—what it changes is the surface of the response, not the reasoning that determines correctness. Attention attribution on the items where a stimulus flips correctness would test this directly; the released per-item outputs identify those items.

**Limitations.** Valence is argued from lexical semantics rather than measured, so the valence contrast is the weakest result here. This bears on interpretation, not on the audit: if มหัศจรรย์ (*mahatsachan*) is less positive than assumed, the noise floor and the equivalence of the target are unchanged, but the A1–C1 contrast is not yet a test of valence. Human norms would make valence a continuous predictor across the whole bank. We audit *minimal* stimuli—minimality is what makes matched nulls constructible—so longer affective paragraphs may behave differently. Models reach 14B; frontier-scale and API models remain untested. Equivalence is established pooled, not per cell. Decoding is greedy, characterising the modal response rather than the output distribution.

## VIII. Conclusion

We asked whether a minimal affective prompt stimulus does anything that a meaningless string of the same shape would not, using a control the prompt-engineering literature does not run: an empirical null distribution of surface-matched inert stimuli. Across 3 model families and two parallel Thai/English task pairs, nothing tested—ours or the published techniques—beats that distribution in the direction of a gain; 6 of 10 stimuli, the target included, are certified inert at $\pm 1.5$ points; and the one stimulus that clears the envelope does harm, through a language switch in one model. The prompt-perturbation noise floor is 2.18 accuracy points, and it is the number we would most like carried forward: it should accompany any future claim of a prompting gain. Code, stimulus bank and per-item generations are released.

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